Cisco Switch Configuration — Command Cheat Sheet

All commands start from `enable` → `configure terminal` unless noted.

```
Switch> enable
Switch# configure terminal
Switch(config)#
```

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1. Hostname

```
Switch(config)# hostname SW1
```

2. Banners

```
SW1(config)# banner motd #Authorized Access Only#
SW1(config)# banner login #Please enter your credentials#
SW1(config)# banner exec #Welcome to SW1 - Logged in successfully#
```

> `#` is the delimiter — use any character not in your message.

3. Console Line

```
SW1(config)# line console 0
SW1(config-line)# password Cisco123
SW1(config-line)# login
SW1(config-line)# exec-timeout 5 0
SW1(config-line)# exit
```

4. Enable Password / Secret

```
SW1(config)# enable password Cisco123        ! legacy, avoid using with secret
SW1(config)# enable secret StrongPass1        ! preferred, encrypted, overrides enable password
SW1(config)# service password-encryption
```

5. VTY Lines (Telnet, before SSH)

```
SW1(config)# line vty 0 15
SW1(config-line)# password Cisco123
SW1(config-line)# login
SW1(config-line)# exit
```

6. Virtual Interface (SVI) for Management

```
SW1(config)# interface vlan 1
SW1(config-if)# ip address 192.168.1.2 255.255.255.0
SW1(config-if)# no shutdown
SW1(config-if)# exit
SW1(config)# ip default-gateway 192.168.1.1
```

7. Prerequisites for SSH

```
SW1(config)# ip domain-name mycompany.local
SW1(config)# username admin privilege 15 secret AdminPass1
SW1(config)# crypto key generate rsa
! How many bits in the modulus [512]: 1024
```

8. SSH Setup

```
SW1(config)# ip ssh version 2
SW1(config)# line vty 0 15
SW1(config-line)# login local
SW1(config-line)# transport input ssh
SW1(config-line)# exit
SW1(config)# ip ssh time-out 60
SW1(config)# ip ssh authentication-retries 3
```

Connect from a client:

```
PC> ssh -l admin 192.168.1.2
```

9. VLAN Creation, Addition, Deletion

Create + name:

```
SW1(config)# vlan 10
SW1(config-vlan)# name Sales
SW1(config)# vlan 20
SW1(config-vlan)# name IT
SW1(config)# vlan 30
SW1(config-vlan)# name Guest
SW1(config-vlan)# exit
```

Add (assign) ports to a VLAN:

```
SW1(config)# interface fastEthernet 0/1
SW1(config-if)# switchport mode access
SW1(config-if)# switchport access vlan 10
SW1(config-if)# exit
```

Assign a range of ports at once:

```
SW1(config)# interface range fastEthernet 0/2 - 5
SW1(config-if-range)# switchport mode access
SW1(config-if-range)# switchport access vlan 20
SW1(config-if-range)# exit
```

Delete a VLAN:

```
SW1(config)# no vlan 30
```

Verify:

```
SW1# show vlan brief
SW1# show vlan id 10
```

10. Trunking

```
SW1(config)# interface fastEthernet 0/24
SW1(config-if)# switchport mode trunk
SW1(config-if)# switchport trunk native vlan 99
SW1(config-if)# switchport trunk allowed vlan 10,20,30
SW1(config-if)# exit
```

11. Inter-VLAN Routing — Router-on-a-Stick

On the switch (trunk port toward router):

```
SW1(config)# interface fastEthernet 0/1
SW1(config-if)# switchport mode trunk
SW1(config-if)# switchport trunk allowed vlan 10,20,30
SW1(config-if)# exit
```

On the router (subinterfaces):

```
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# no shutdown
Router(config-if)# exit

Router(config)# interface gigabitEthernet 0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config-subif)# exit

Router(config)# interface gigabitEthernet 0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
Router(config-subif)# exit
```

Verify:

```
Router# show ip interface brief
Router# show interfaces gigabitEthernet 0/0.10
```

12. Inter-VLAN Routing — Layer 3 Switching (SVI-based)

On a multilayer (Layer 3) switch:

```
SW1(config)# ip routing
SW1(config)# vlan 10
SW1(config-vlan)# exit
SW1(config)# interface vlan 10
SW1(config-if)# ip address 192.168.10.1 255.255.255.0
SW1(config-if)# no shutdown
SW1(config-if)# exit

SW1(config)# vlan 20
SW1(config-vlan)# exit
SW1(config)# interface vlan 20
SW1(config-if)# ip address 192.168.20.1 255.255.255.0
SW1(config-if)# no shutdown
SW1(config-if)# exit
```

If the switch connects to other routers/networks, also make the port routed (not switched):

```
SW1(config)# interface fastEthernet 0/24
SW1(config-if)# no switchport
SW1(config-if)# ip address 10.0.0.1 255.255.255.252
```

Verify:

```
SW1# show ip route
SW1# show ip interface brief
SW1# show vlan brief
```

13. Verification & Service Confirmation

MAC address table:

```
SW1# show mac address-table
SW1# show mac address-table dynamic
SW1# show mac address-table interface fastEthernet 0/1
SW1# show mac address-table vlan 10
SW1# clear mac address-table dynamic
```

Interface / port status:

```
SW1# show interfaces status
SW1# show interfaces fastEthernet 0/1
SW1# show interfaces fastEthernet 0/1 switchport
SW1# show ip interface brief
```

Trunk-specific:

```
SW1# show interfaces trunk
```

VLAN verification:

```
SW1# show vlan brief
SW1# show vlan id 10
```

Connectivity testing (ping / traceroute):

```
SW1# ping 192.168.10.1
SW1# ping 192.168.20.5
PC> ping 192.168.10.1
PC> tracert 192.168.20.5          ! Windows
Router# traceroute 192.168.20.5   ! Cisco IOS
```

SSH / remote access confirmation:

```
SW1# show ip ssh
SW1# show ssh
SW1# show line vty 0 15
PC> ssh -l admin 192.168.1.2
PC> telnet 192.168.1.2            ! should fail if SSH-only is configured
```

Neighbor / topology confirmation:

```
SW1# show cdp neighbors
SW1# show cdp neighbors detail
SW1# show lldp neighbors
```

Port security (if configured):

```
SW1# show port-security
SW1# show port-security interface fastEthernet 0/1
```

Spanning-tree (if multiple switches):

```
SW1# show spanning-tree
SW1# show spanning-tree vlan 10
SW1# show spanning-tree summary
```

Routing table (router or L3 switch):

```
Router# show ip route
SW1# show ip route
```

Save and back up config:

```
SW1# copy running-config startup-config
SW1# write memory                     ! shorthand for the above
SW1# show running-config
SW1# copy running-config tftp         ! backup to TFTP server, if available
```

Quick Reference Table

Task	Command
Hostname	`hostname <name>`
MOTD banner	`banner motd #text#`
Console password	`line console 0` → `password <pwd>` → `login`
Enable secret	`enable secret <pwd>`
VTY password	`line vty 0 15` → `password <pwd>` → `login`
SVI IP	`interface vlan 1` → `ip address <ip> <mask>`
Default gateway	`ip default-gateway <ip>`
Domain name	`ip domain-name <name>`
Local user	`username <user> privilege 15 secret <pwd>`
RSA keys	`crypto key generate rsa`
SSH only on VTY	`transport input ssh`
Create VLAN	`vlan <id>` → `name <name>`
Assign port to VLAN	`switchport mode access` → `switchport access vlan <id>`
Delete VLAN	`no vlan <id>`
Trunk port	`switchport mode trunk`
Router subinterface	`interface g0/0.10` → `encapsulation dot1Q 10`
Subinterface IP	`ip address <ip> <mask>` (inside subinterface)
Enable L3 routing on switch	`ip routing`
SVI as gateway	`interface vlan <id>` → `ip address <ip> <mask>`
Routed switch port	`no switchport` → `ip address <ip> <mask>`
MAC address table	`show mac address-table`
Clear MAC table	`clear mac address-table dynamic`
Interface status	`show interfaces status`
Trunk details	`show interfaces trunk`
Ping	`ping <ip>`
Traceroute	`traceroute <ip>` (router) / `tracert <ip>` (Windows PC)
SSH status	`show ip ssh`
CDP neighbors	`show cdp neighbors`
Port security status	`show port-security`
Spanning-tree status	`show spanning-tree`
Routing table	`show ip route`
Save config	`copy running-config startup-config`